

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze various obvious and hidden quality costs of a firm for quality system economics.
2. Apply various quality control tools for troubleshooting to reduce sporadic quality problems.
3. Conduct process capability analysis
4. Perform line and angular measurements using appropriate metrological instruments.

Text Books

5. Juran, J.M. and Gryna, F.M., Quality Planning & Analysis, 3rd Edition, McGraw Hill (2001).
6. Grant, E.L., Statistical Quality Control, 3rd Edition, McGraw Hill (2014).
7. Beckwith, T.G., Marangoni, R.D., and Lienhard, J.H., Mechanical Measurements, 6th Edition, Pearson Education (2006).

Reference Books

1. Feignbaum, A.V., Total Quality Control, 3rd Edition, McGraw Hill (1991).
2. Raghavendra, N.V., and Krishnamurthy, L., Engineering Metrology and Measurements, 2nd Edition, Oxford University Press (2013).

Evaluation Scheme

S.No	Assessment	Weightage (%)
4.	MST	30
5.	EST	45
6.	Sessional (tutorial assignments/quizzes)	25

UME626 : Process Engineering Standards

L	T	P	Cr
3	0	0	3.0

Course Objective: To introduce the concepts of design codes/standards and engineering specification for process engineering. To educate on the process of use of design standard to accomplish process engineering design. To impart knowledge on the method of development of flow diagram for reliable and safe operation of process plant. To educate on the process of use of component selection using manufactures' catalogue and satisfying other constraints and criteria in process engineering.

Syllabus

Design Standards & Engineering Specification: Introduction to industrial utility systems, such as steam generators, heat exchangers, pumps and compressed air systems, introduction to different design standards and codes, such as IS, DIN, BS, ISO, ASME, API, Hydraulic Institute Standards etc. and applicability, need for standards in professional design and engineering, development & use of engineering specifications, data sheet, BoM etc.

Process Equipment Design: Use of standards/codes in typical mechanical and process equipment design, as example pressure vessel design, boiler tube design, shell and tube heat exchanger design, etc. – classification of pressure vessel, mechanical design of shell, design against internal and external pressure, selection of heads and joints, boiler and heat exchanger pipe thickness calculation and material selection.

Process Flow Development: Fundamentals of P&ID, anatomy, general rules, and common industrial symbols, legends and tagging for valves, fittings, actuators, instruments etc., development of system P&IDs/process flow for different industrial system, as example industrial utility systems - pumping network, compressed air systems, material handling etc.

Component Selection: Equipment/component selection in practical scenario - optimizing design, cost, spares, maintenance, space, availability and other constraints and criteria, use of manufacturers' catalogue to select commonly used components, as example for motor, conveyor belt, coupling, pump, compressor etc.

Project:

A project is included in evaluation which will cover developing a complete piping network/P&ID and pump/compressor selection for a process plant including selection of pump/compressor model, design of pressure vessel with instruments and safety attachment, pipe thickness calculation, instrument model selection with specification of each item.

Course Learning Objectives (CLO)

The students will be able to:

1. Develop engineering specifications and data sheets for process plants
2. Design process equipment using design standards/codes
3. Develop Process and Instrumentation Diagram (P&ID)